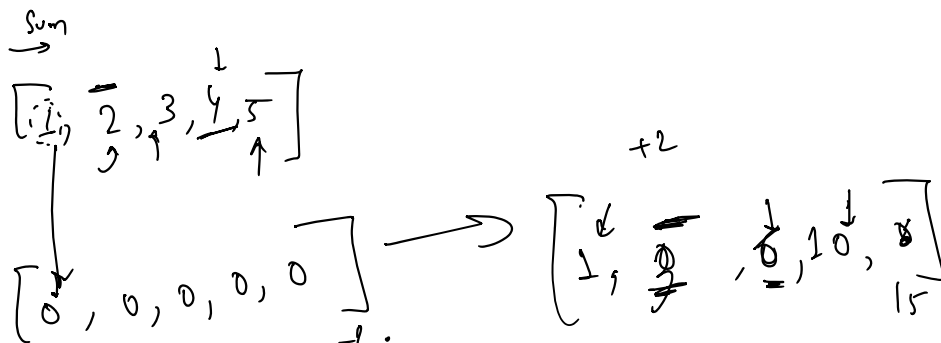
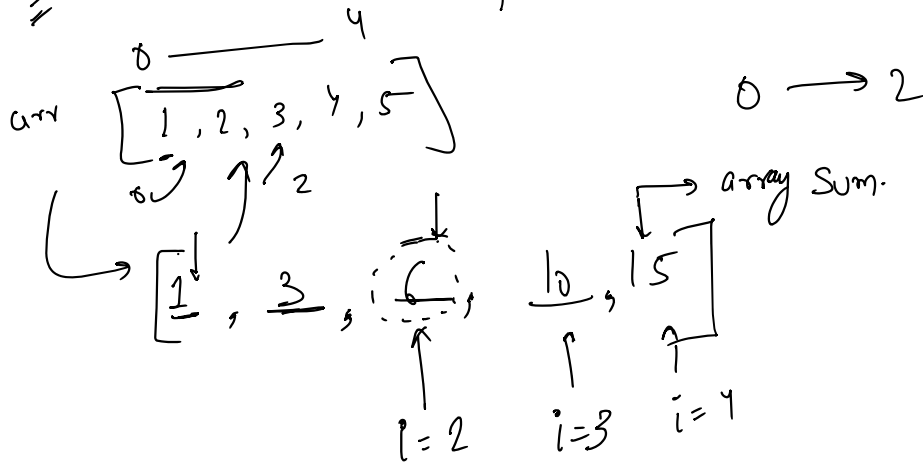


Arrays 9

17 April 2026 13:30

Q: Find & store the prefix sum of an array



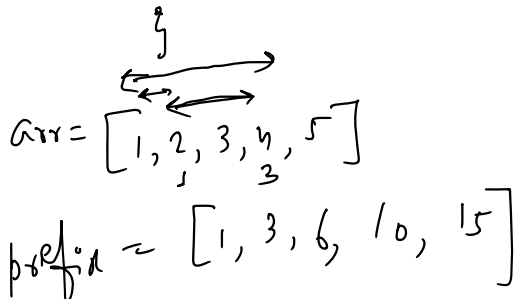
```
int[] arr = {1, 2, 3, 4, 5};
int n = arr.length;
```

```
int[] prefix = new int[n]; // 0
```

```
prefix[0] = arr[0];
```

```
for (int i = 1; i < n; i++) {
```

```
    prefix[i] = prefix[i-1] + arr[i];
```



```
int i = 1 → 3
sum += arr[i];
```

i, j → find the sum from i to j.

$prefix[i] = prefix[i-1]$

if ($i > 0$)

$i=0$, return $prefix[i]$

$i=0$, $j=2$

// Suffix array

int[] arr = {1, 2, 3, 4, 5}

↓ ↑ ←

for ($i=n-2 \rightarrow 0$) {

$Suffix[i] = Suffix[i+1] + arr[i];$

}

// left maximum array

↓ [3, 2, 1, 6, 4, 2]

[3, 3, 3, 6, 6, 6]

arr = [3, 2, 1, 6, 4, 2]

↓ ↓ ↓ ↓ ↓

arr = [3, 4, 7, 3, 2, 6]

left_max = [0, 0, 0, 0, 0, 0] → [3, 4, 7, 3, 2, 6]

```
int n = arr.length;
```

```
int[] left_max = new int[n];
```

```
left_max[0] = arr[0];
```

```
for (int i = 1; i < n; i++) {
```

```
    left_max[i] = max(left_max[i-1], arr[i]);
```

```
}
```

[3, 4, 7, 3, 2, 6]

left_max [3, 4, 7, 7, 7, 7]

// right_max

```
int n = arr.length;
```

```
right_max = arr[n-1];
```

```
for (int i = n-2; i >= 0; i--) {
```

```
    right_max = max(right_max, arr[i]);
```

```

for(int i = n-2; i >= 0; i--)
    right_max[i] = max(right_max[i+1], arr[i]);
}

```